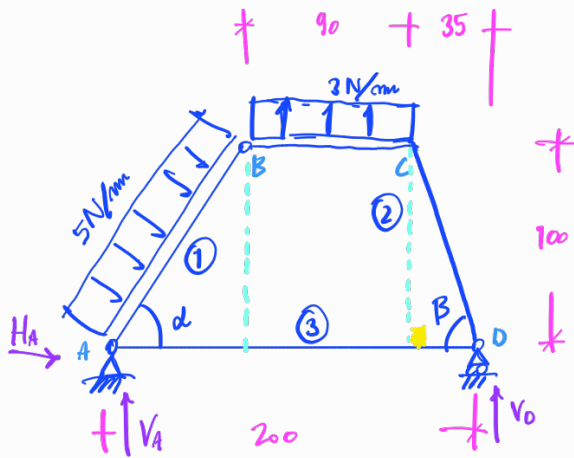




$$AB = \sqrt{100^2 + (200 - 90 - 35)^2} = 125 \text{ mm}$$

$$CD = \sqrt{100^2 + 35^2} = 105,95 \text{ mm}$$

$$\alpha = \arctg\left(\frac{100}{75}\right) = 53,13^\circ$$

$$\beta = \arctg\left(\frac{100}{35}\right) = 70,71^\circ$$

$$\sin(\alpha) = 0,8$$

$$\cos(\alpha) = 0,6$$

$$\sin(\beta) = 0,94$$

$$\cos(\beta) = 0,33$$

Equazioni cardinali:

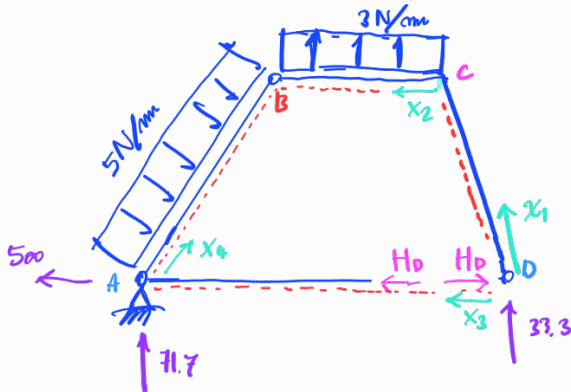
$$\sum H = 0 \quad H_A + 5 \cdot 125 \cdot 0,8 \sin 53,13 = 0 \quad \rightarrow \quad H_A = -500 \text{ N}$$

$$\sum V = 0 \quad V_A + V_D + 3 \cdot 90 - 5 \cdot 125 \cos 53,13 = 0 \quad \rightarrow \quad V_A = 71,7 \text{ N}$$

$$\sum M_A = 0 \quad -V_D \cdot 200 - 3 \cdot 90 \left(75 + \frac{90}{2}\right) + 5 \left(\frac{125^2}{2}\right) = 0 \quad \rightarrow \quad V_D = 33,3 \text{ N}$$

$$-H_D \cdot 100 - 33,3 \left(90 + 35\right) - 3 \cdot \frac{90^2}{2} = 0$$

$$H_D = -163,1 \text{ N}$$



Calcolo delle azioni interne:

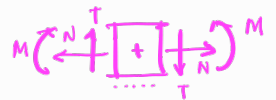
TRATTO DA $0 \leq x_3 \leq 200$



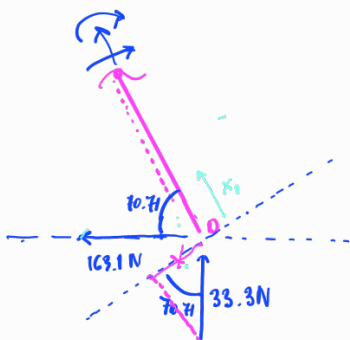
$$N_{DA} = 163,1 \text{ N}$$

$$T_{DA} = 0$$

$$M_{DA} = 0$$



TRATTO DC



$$N_{DC} + 163,1 \cdot \cos 70,71 + 33,3 \cdot \sin 70,71 = 0$$

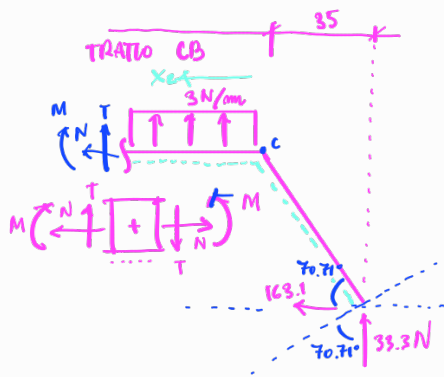
$$N_{DC} = -85,31 \text{ N}$$

$$T_{DC} + 33,3 \cos 70,71 - 163,1 \sin 70,71 = 0$$

$$T_{DC} = 143 \text{ N}$$

$$M_{DC} - 33,3 \cos 70,71 x_1 + 163,1 \sin 70,71 x_1 = 0$$

$$M_{DC} = -143 x_1 \quad \begin{cases} M_{DC}(x_1 = 0) = 0 \\ M_{DC}(x_1 = 105,95) = -15150,85 \text{ Nmm} \end{cases}$$



$$N_{CB} = -163.1 \text{ N}$$

$$T_{CB} + 33.3 + 5 \cdot X_2 = 0$$

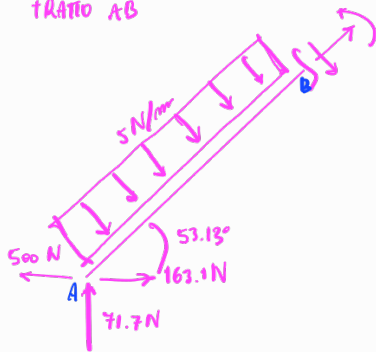
$$T_{CB} = -3X_2 - 33.3$$

$$\begin{cases} X_2 = 0 & T_{CB} = -33.3 \text{ N} \\ X_2 = 90 & T_{CB} = -303.3 \text{ N} \end{cases}$$

$$M_{CB} - 33.3(X_2 + 35) + 163.1 \cdot 100 - 5 \cdot \frac{X_2^2}{2} = 0$$

$$M_{CB} = 1.5 X_2^2 + 33.3 X_2 - 15147.15 = \begin{cases} M_{CB}(X_2 = 0) = -15147.15 \\ M_{CB}(X_2 = 90) = 0 \end{cases}$$

TRATIO AB



$$N_{AB} - 500 \cos 53.13 + 71.7 \cdot \sin 53.13 + 163.1 \cos 53.13 = 0$$

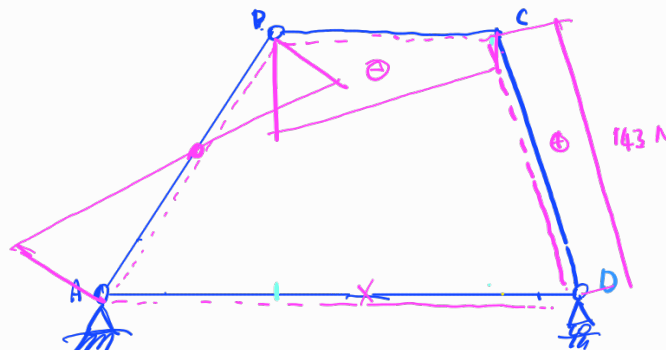
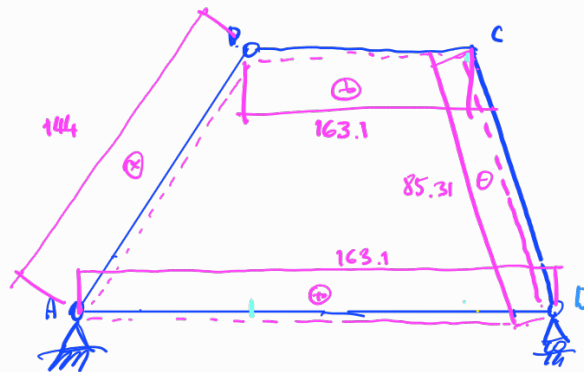
$$N_{AB} = 144.78 \text{ N}$$

$$T_{AB} - 500 \cdot \sin 53.13 - 71.7 \cos 53.13 + 163.1 \sin 53.13 + 5 X_3 = 0$$

$$T_{AB} = \begin{cases} T_{AB}(X_3 = 0) = 312.5 \text{ N} \\ T_{AB}(X_3 = 125) = -312.5 \text{ N} \end{cases}$$

$$M_{AB} - 500 \cdot \sin 53.13 \cdot X_3 - 71.7 \cos 53.13 \cdot X_3 + 163.1 \sin 53.13 \cdot X_3 + 5 \cdot \frac{X_3^2}{2} = 0$$

$$M_{AB} = \begin{cases} M_{AB}(X_3 = 0) = 0 \\ M_{AB}(X_3 = 125) = 0 \end{cases}$$



M

⊖

⊕

